

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CSE313: DESIGN OF LANGUAGE PROCESSOR

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Theory of Computation

A. Outline of the Course:

Sr. No.	Title of the unit	Number of hours
1	Overview of Language Processors & Lexical Analysis	7
2	Syntax Analysis	5
3.	Parsing Methods	16
4.	Syntax-Directed Translation & Intermediate Code Generation	9
5.	Runtime Environment & Code Generation	8

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

Unit No.	Unit/Topics	Hours	Weightage (%)
1	Overview of Language Processors & Lexical Analysis	07	15%
	Language Processors, The Structure of a Compiler Application of Compiler Technology Lexical Analysis: The Role of Lexical Analyzer, Specification of Tokens, Recognition of Tokens, Lexical Analyzer Generator LEX.		

2	Syntax Analysis	05	5%
	Role of the Parser, Representative Grammar, Syntax Error Handling, Error-recovery Strategies.		
3.	Parsing Methods	16	35%
	Top Down Parsing: Recursive-Descent, Parsing, FIRST and FOLLOW, LL (1) grammar, Non-recursive Predictive Parsing, Construction of Non-recursive Predictive Parsing Table, Error Recovery in Predictive Parsing. Bottom-up Parsing: Shift-Reduce Parsing, Conflicts during Shift-Reduce Parsing, Introduction to LR Parsing, L-R Parsing Algorithm, Viable Prefixes, Simple LR Parser (SLR), Construction of Simple LR Parsing Table, Canonical LR(1), Construction of LR(1) Parsing Table, Look Ahead LR (LALR), Construction of LALR Parsing Table, Parser Generator – Yacc.		
4.	Syntax-Directed, Translation & Intermediate Code Generator	09	30%
	Syntax-Directed Definitions, Dependency Graphs, S-attributed Definitions, L-attributed Definitions, Application of Syntax Directed Translation, Syntax Directed Translation Schemes. Intermediate Code Generation: Variants of Syntax Trees, Three Address Code, Control Flow.		
5.	Runtime Environment & Code Generation	08	15%
	Storage Organization, Activation Trees, Activation Records, Calling Sequence, Heap Management, Introduction to Garbage Collection. Code Generation: Issues in Code Generator, The Target Language, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, A simple Code Generator, Peephole Optimization.		

C. Course Outcome:

After completion of the course, Students will be able to

CO1	Understand the fundamentals of the compiler identify the relationships among different phases of the compiler and use the knowledge of the Lex tool
CO2	Describe the Role of Parser and the various error recovery strategies
CO3	Develop the parsers and experiment with the knowledge of different parser designs.
CO4	Develop a semantic analysis scheme to generate intermediate code
CO5	Summarize various optimization techniques used for dataflow analysis and generate machine code from the source code

D. Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	2	-	2	-	-	-	1	-	-	-	-	1
CO2	-	2	-	1	-	2	-	1	2	-	-	-	2	-
CO3	-	-	3	1	1	-	-	1	-	1	-	-	1	-
CO4	-	-	3	-	1	2	-	-	1	-	-	-	-	1
CO5	-	1	-	2	-	1	-	2	-	3	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put "-".

E. Recommended Study Material:

❖ Text Books:

1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, "Compilers Principles, Techniques and- Tools", Pearson Education Asia.
2. M. Dhamdhere, "System Programming and Operating Systems", Tata McGraw-Hill.
3. Steven S. Muchnick. Advanced Compiler Design and Implementation

❖ Reference Books:

1. Allen I. Holub "Compiler Design in C", Prentice Hall of India.
2. C. N. Fischer and R. J. LeBlanc, "Crafting a compiler with C", Benjamin Cummings.
3. J.P. Bennet, "Introduction to Compiler Techniques", Second Edition, Tata McGraw-Hill
4. Henk Alblas and Albert Nymeyer, "Practice and Principles of Compiler Building with C", PHI.

5. Kenneth C. Louden, "Compiler Construction: Principles and Practice", Thompson Learning.

❖ **Web material:**

1. <http://compilers.iecc.com/crenshaw>
2. <http://www.compilerconnection.com>

❖ **Software:**

1. LEX & YACC